# Computing Infrastructure (Linux essentials)
# SWE4303

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## Assessment Brief

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|---|---|
| Assessment Number | 002 |
| Assessment Type (and weighting) | Portfolio (60%) |
| Assessment Name | Portfolio of tasks related to developing scripted solutions, based on the client/server model. |
| Assessment Submission Date | 15/01/2026 |

Learning Outcomes Assessed:

LO3: Formulate scripted solutions to automate operating system services

LO4: Implement practical solutions based on the client/server model to solve a given problem

Use of Generative Artificial Intelligence (GAI) Applications in this Assessment

| AI Status | Application | Notes |
|---|---|---|
| Category C | GAI is an integral part of the assessment and should be used. | Grammar and/or spell checkers may be used to correct individual words and sentences.<br><br>Your skills in using and evaluating GAI generated content is part of the assessment task in question. GAI generated content should be clearly identified and acknowledged.<br><br>Any GAI generated content which is presented as your own original work and is not acknowledged will be assessed for academic misconduct. |

Assessment Task:

In a multinational company which uses client/server model based on a Linux operating system, the system administrator consumes a lot of time for his daily administrative tasks. So, he assigns you to write some automated scripts which will solve the problem.

Declaration: At the end of the assessment, you should also include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment.

Level HE4 - It is expected that the Reference List will contain between five and ten sources. As a MINIMUM the Reference List should include one refereed academic journal and three academic books

## Assessment Tasks

- **Task 1 — System Preparation**
- **Task 2 — User & Group Automation**
- **Task 3 — Backup & File System Administration**
- **Task 4 — System Monitoring Automation**
- **Task 5 — Network Health Checks**
- **Task 6 — Bash Customization**
- **Task 7 — Admin Toolkit Menu**
- **Task 8 — Documentation**

- **Task 9 — Final Presentation (Week 13)**
  You will deliver a **5–8minute presentation** demonstrating:
- Your scripts
- How they work
- What tasks they automate
- Your admin menu
- Improvements you would make with more time